

Vishal Khoje

Principal Engineer | Engineering Manager | Solution Architect

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SUMMARY

Solution Architect with **13+ years of experience** building AI-powered, cloud-native distributed systems, specializing in LLMs, RAG, and agent-based architectures at scale. Delivered 10x performance gains, 70%+ productivity improvements, and highly reliable (99.9%+) platforms.

Leads end-to-end GenAI systems, including retrieval pipelines and cost - latency optimization, with a strong foundation in distributed architectures.

SKILLS

- **Architecture & Leadership:** Cloud Architecture, Solution Design, AI Architecture, Distributed Systems, Microservices, Event-Driven Architecture, Technical Leadership, Scalability
- **Tech Stack:** Azure, AWS, LangChain, LangSmith, RAG, Agentic AI, OpenAI SDK, MCP, CrewAI, Pinecone, Kubernetes, Docker, Kafka, React, Python, FastAPI, PostgreSQL, Node.js, JavaScript, React.js

PROFESSIONAL EXPERIENCE

Mathco (TheMathCompany), Bangalore

Feb 2023–Present

Platform Solution Architect

- **AI System Architecture & Production Experience**
 - Owned the retrieval layer for a customer-facing AI agent handling 20,000+ daily queries, including architecture design, cost - latency optimization, and incident management.
 - Improved retrieval precision by 31pp by implementing hybrid search (vector + BM25), addressing 18% loss on long-tail queries with only +22ms latency overhead.
 - Designed and deployed a guardrail system after a prompt injection incident that misrouted 0.4% of production queries to unintended tool endpoints, implementing policy enforcement at the I/O boundary to prevent recurrence.
 - Architecting an AI-driven QA automation platform to autonomously generate, execute, and report test cases end-to-end - targeting elimination of manual QA effort and full SDLC test coverage.
 - Delivered an AI-powered insights platform (Azure OpenAI, RAG, LangChain) automating 100+ reports/day. Reduced analyst effort by 70%+, accelerating enterprise insights and decision-making.
- Led end-to-end architecture and client deployment of the NucliOS platform (AWS/ Azure), implementing OAuth2, SSO, rate limiting, DevOps (CI/CD) pipeline and ensuring reliability through observability (Grafana, CloudWatch) and messaging (Azure Service Bus), improving system performance and contributing to higher client onboarding and platform adoption.
- Drove end-to-end presales architecture for 5+ enterprise clients, creating solution proposals and demos that contributed to 20% improvement in deal conversion rate.
- Cut cloud costs by ~25% (~\$300/month) using auto-scaling instead of fixed server configurations. Automated dev/QA scaling (weekend downscale, weekday upscale) for efficient resource utilization.
- Improved NucliOS platform performance by 10x (3s → 300ms) by redesigning architecture, caching, and implementing server-side pagination and message queueing, enabling scalability for 50K+ users.
- Led end-to-end Full-stack development using React.js and Python (FastAPI), owning frontend, backend APIs, and integrations. Delivered scalable, high-performance user experiences for enterprise platforms.
- Built and scaled the engineering team from scratch to 10+ members, mentoring engineers and tech leads to drive high-quality delivery, ownership, and engineering excellence.
- Introduced ADRs (Architecture Decision Records), RFC processes, and automated code quality gates, reducing production incidents by 30% and onboarding time for new engineers.

Stefanini Group, Pune

Technical Lead

Jan 2022–Jan 2023

- Led delivery of a global, multi-region product across 6+ countries (US & Europe), managing a team of 6 engineers and meeting all delivery deadlines across every region without exception.
- Drove end-to-end system modernization (migration, modularization, CI/CD), reducing production bugs and incidents through enforced code quality standards, peer reviews, and automated testing pipelines.
- Owned architecture and engineering leadership across a distributed team - establishing best practices that improved system maintainability, scalability, and release confidence.

UST Xpanxion, Pune

Associate Leader - Software

March 2020–Dec 2021

- Developed reusable SDKs and modules (React, TypeScript, Monorepo) that standardized and accelerated personalization integration across the client's product suite.
- Won a client-side hackathon by building a voice-enabled web experience – selected as the winning solution among competing teams for innovation and technical execution.

Reliance Industries Limited, Navi Mumbai

Full Stack Developer

Oct 2018–March 2020

- Delivered an AI-powered Employee Virtual Assistant for Reliance's EmpXP platform, used by 200K+ employees, reducing HR query resolution time by 50% and improving employee experience.

sumHR (Sutra HR), Navi Mumbai

Senior Software Engineer

April 2016–Sep 2018

- Built and scaled a HRMS & Payroll SaaS platform supporting 500+ organizations, generating multi-million revenue, achieving 99.9% uptime, and leading a team of 5 engineers to deliver high customer satisfaction through Agile execution.

eBrandz Solutions Pvt. Ltd. Mumbai

Senior Software Engineer

Jun 2014–March 2016

- Built and maintained RESTful APIs (PHP) for the Brand360 mobile app, including content aggregation from third-party platforms via web scraping.

KEY PROJECTS

- AI Resume Chatbot https://huggingface.co/spaces/vishalkhoje/career_conversation
 - Built conversational AI system enabling recruiters to interact with candidate profiles
 - Built interactive AI career assistance with agentic workflows, evaluation, monitoring, and self-improving capabilities for continuous optimization.
- AI QA Automation Platform (In Development)
 - Designing AI-driven workflows to automate the full test lifecycle
 - Targeting elimination of manual QA effort and end-to-end SDLC coverage

AI LEARNING & CREDENTIALS

- AWS Certified AI Practitioner AIF-C01 - Completion Badge (2024)
- AI Agents Fundamentals – (Hugging Face)
- Microsoft Learn AI Skills Challenge - Certificate of Completion

HONORS & AWARDS

- Innovation & Creativity Award - TheMathCompany (Mathco), Feb 2024
- Hacktoberfest 2023 Honored Contributor - Issued by Hacktoberfest · Oct 2023

EDUCATION

MASTER OF COMPUTER APPLICATION (ENGG.),
PUNE UNIVERSITY

April 2013
